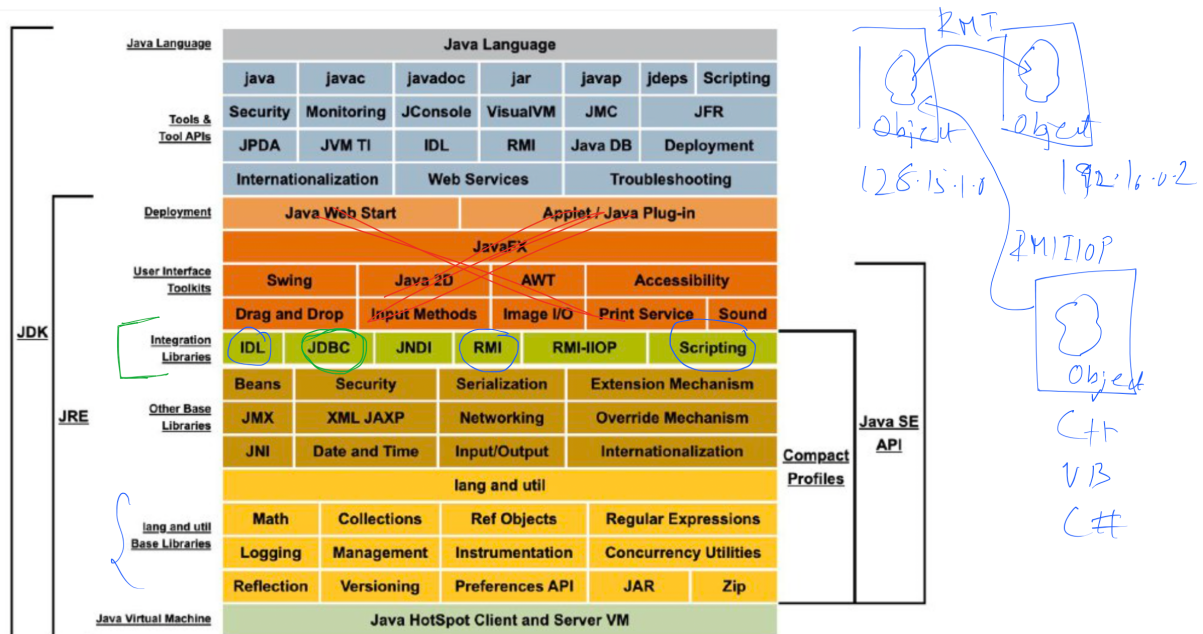
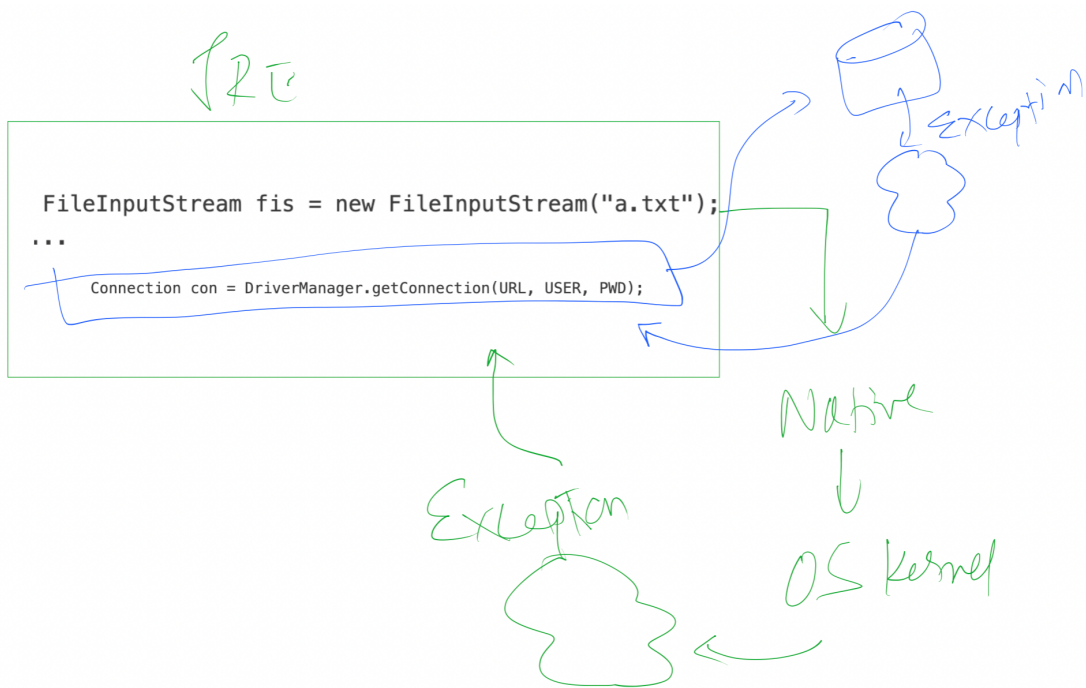
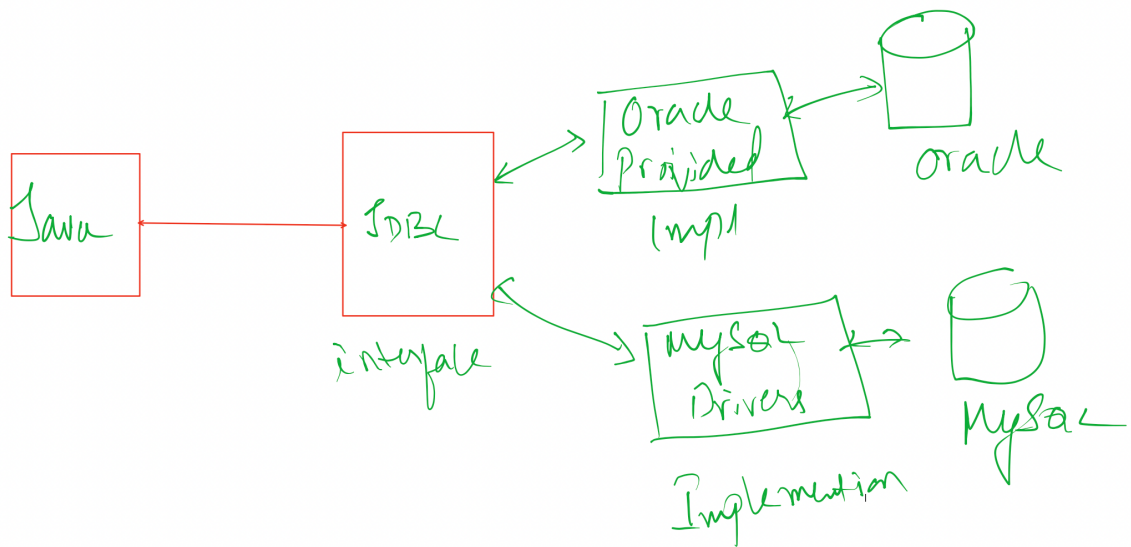




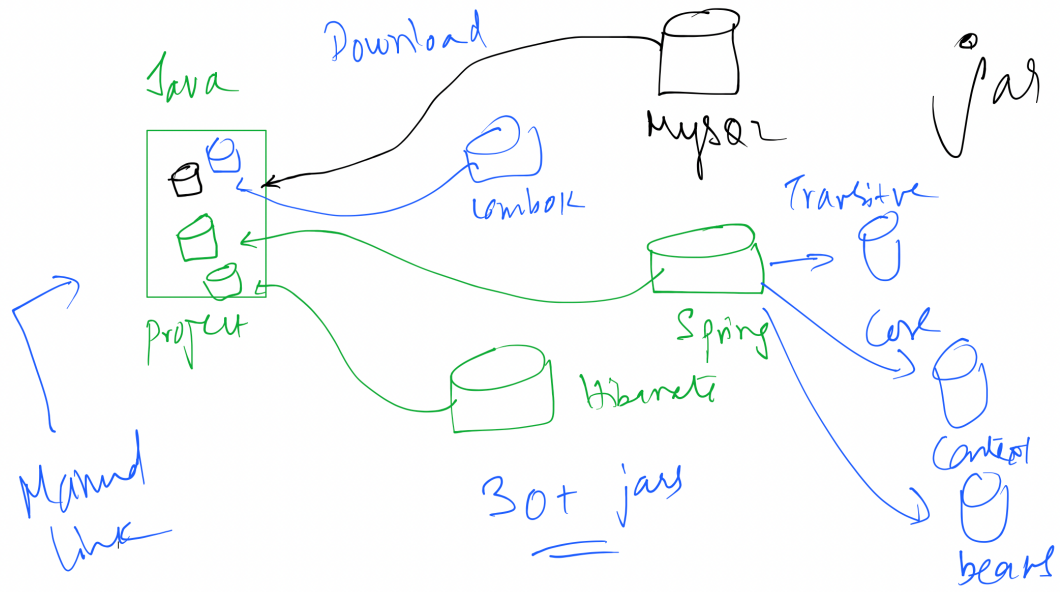


ResultSet object

	Col - 1	Col - 2	Col - 3
Row-1			
Row-2			
Row-3			

ResultSet pointer

Before Maven:

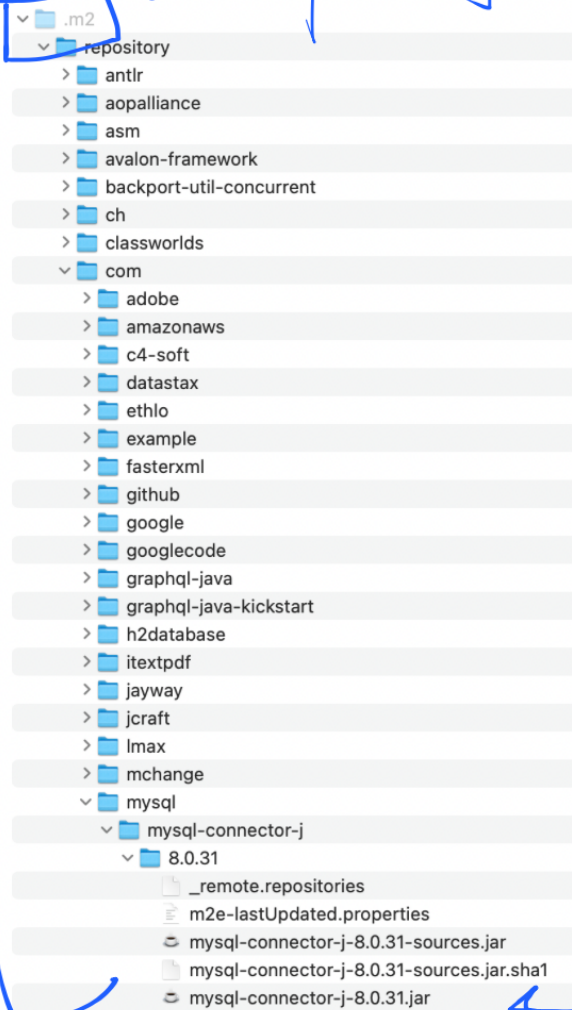


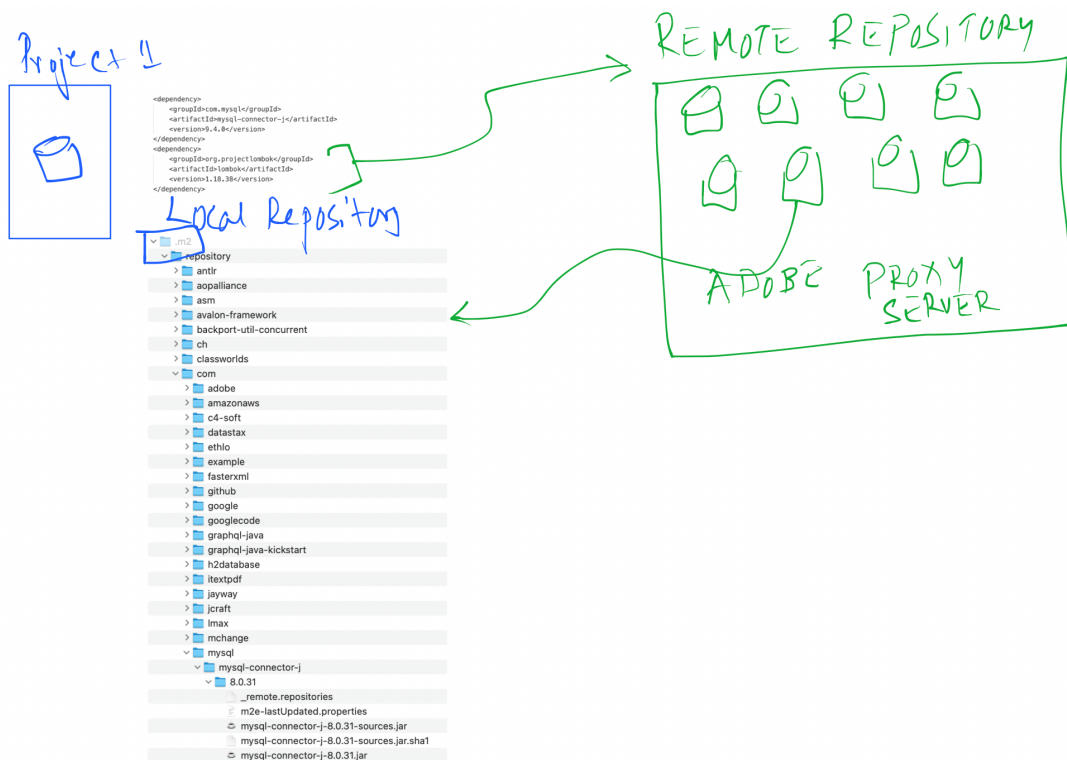
Project 1



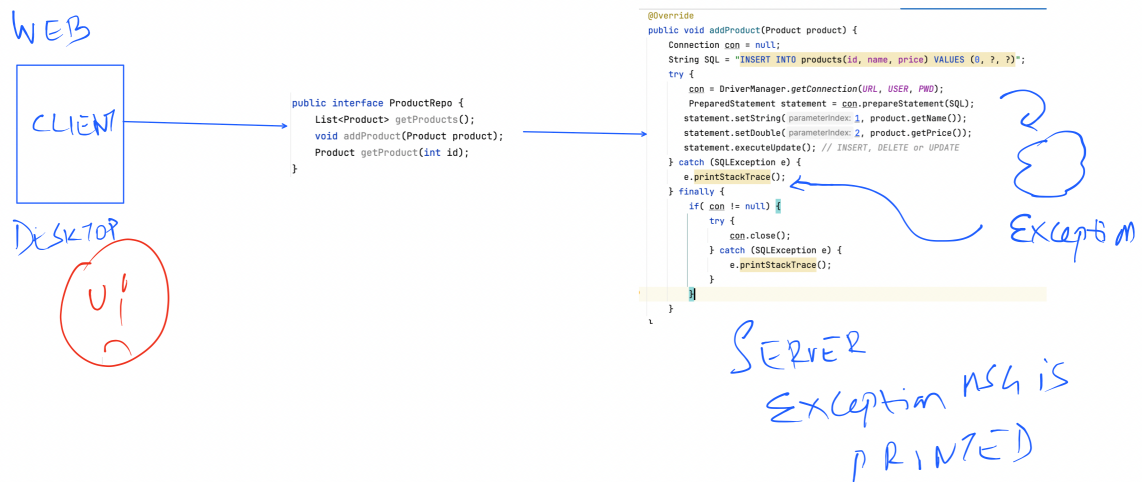
```
<dependency>
  <groupId>com.mysql</groupId>
  <artifactId>mysql-connector-j</artifactId>
  <version>9.4.0</version>
</dependency>
<dependency>
  <groupId>org.projectlombok</groupId>
  <artifactId>lombok</artifactId>
  <version>1.18.38</version>
</dependency>
```

Local Repository

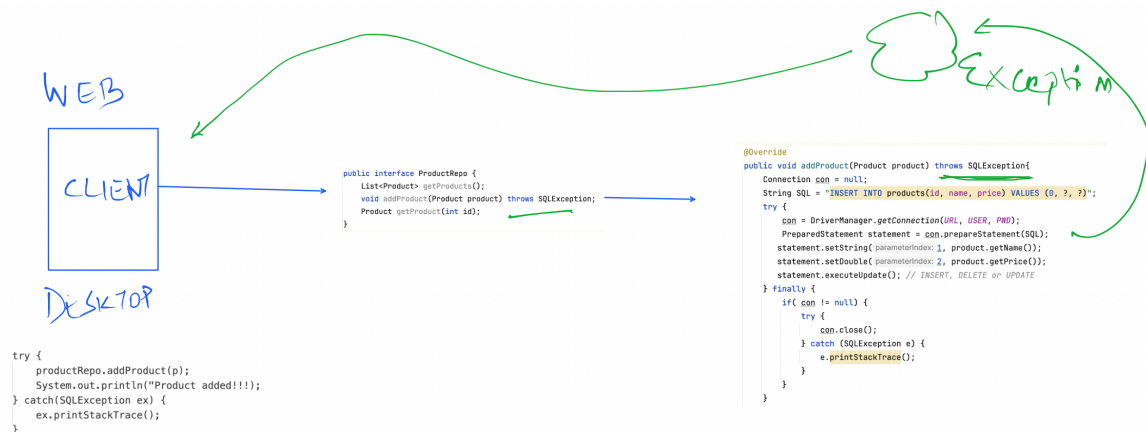




Exception Handling Scenario 1: Client has no clue, because exceptions are handled in the backend

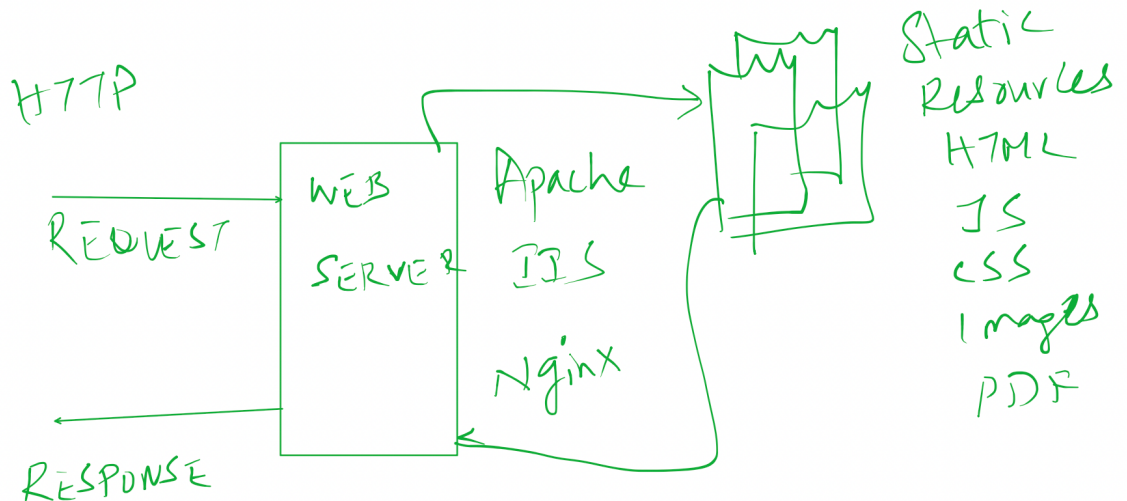


Exception Handling Scenario 2:



Issues:

- 1) Tightly coupled to SQL, no abstraction; changing implementation to MongoDB / Casandra ... needs interface and client code to change.
- 2) Exception messages gives away complete details to client like table name, column names ,...
- 3) Exception messages are complex for client to UNDERSTAND, like ORA9001: Unique constraint violation exceptions



HTTP

REQUEST

RESPONSE

WEB
SERVER

Dynamic Content
Engine | Application
Container

SERVLET CONTAINER

PHP

NODE JS

• .NET

Tomcat / Jetty / Netty

<http://localhost:8080/login>

Thread pools

request

response

DI

LoginServlet
/login

RegisterServlet
/register

Servlet Container

war
WEB ARCHIVE

```
database
|
| HTML / CSS / JS
|
| WEB-INF
|   web.xml
|   classes
|     LoginServlet.class
|     RegisterServlet.class
|     ProductServlet.class
```

Install